

PeakAndPack Test Cases (Manual)

Phase: 3, Approach A, Manual, No AI **Based on:** `test-strategy-manual.md` (Phase 2) and `peakandpack-requirements.md` (Phase 1) **Author:** RN

These test cases were written by hand, working through happy path, negative, and edge cases for each feature area. The riskiest areas from the Phase 2 risk ranking (Checkout, Auth/Orders) get the most coverage. TC-IDs are grouped by area in blocks of ten.

A handful of these are written to fail on purpose, because the expected result is the correct behaviour per the requirements, and the current build has known issues. Those are called out in the Notes column.

Products (TC-001 to TC-010)

TC-ID	Title	Steps	Expected	Priority
TC-001	List all products	GET the product list with no filters	200, array of products with id, name, price	High
TC-002	Product has valid price	Inspect any product's price field	Price is a positive number, two decimal places	High
TC-003	Single product by id	Request a product by a valid id	200, the matching product object	Medium
TC-004	Product by missing id	Request a product id that does not exist	404, clear not-found message	Medium
TC-005	Product price boundary	Inspect a product priced at exactly 0.00	Either rejected at creation, or clearly marked free, never a silent negative total later	Medium
TC-006	Special characters in name	View a product whose name has an apostrophe or accent	Renders correctly, no broken encoding	Low

Search and sort (TC-011 to TC-020)

TC-ID	Title	Steps	Expected	Priority
TC-011	Search by keyword	Search for a word in a product name	200, only matching products returned	Medium
TC-012	Search with no results	Search for a term that matches nothing	200, empty list, not an error	Medium
TC-013	Search with empty query	Submit search with a blank query	400 or full list per spec, never a 500	High
TC-014	Search with special characters	Search using quotes, %, or a script tag	Handled safely, no injection, no crash	High
TC-015	Sort by price ascending	Sort the list low to high	Products ordered by price ascending	Low
TC-016	Sort by invalid field	Sort by a field that does not exist	Falls back to default or 400, never a crash	Low

Cart (TC-021 to TC-035)

TC-ID	Title	Steps	Expected	Priority
TC-021	Add item to cart	Add one in-stock product	Item in cart, total updates correctly	High
TC-022	Add multiple quantities	Set quantity to 3 for one item	Line total is price times three	High
TC-023	Remove item from cart	Remove an item from a cart with two items	Item gone, total recalculated	High
TC-024	Update quantity to zero	Set an item's quantity to 0	Item removed or rejected, never a zero-priced ghost line	Medium
TC-025	Negative quantity	Set quantity to -1	Rejected, cart unchanged	High
TC-026	Very large quantity	Set quantity to 99999	Handled or capped per spec, no overflow in total	Medium
TC-027	Cart total accuracy	Add three different items	Cart total equals the sum of line totals	Critical
TC-028	Cart persists in session	Add item, reload, check cart	Item still present for the same session	Medium
TC-029	Empty cart total	View an empty cart	Total is 0.00, no error	Low

Auth (TC-036 to TC-050)

TC-ID	Title	Steps	Expected	Priority
TC-036	Register new user	Register with valid email and password	201, account created	Critical
TC-037	Register duplicate email	Register with an email already in use	409 or clear error, no second account	High
TC-038	Register with no password	Submit registration with password blank	400, account not created	Critical
TC-039	Register weak password	Submit a one-character password	Rejected per password policy in spec	High
TC-040	Login valid credentials	Log in with a correct email and password	200, session or token issued	Critical
TC-041	Login wrong password	Log in with a correct email, wrong password	401, no session issued	Critical

TC-042	Login wrong password	Log in with a correct email, wrong password	401, no session issued	Critical
TC-042	Login unknown email	Log in with an email never registered	401, same generic error as wrong password	High
TC-043	Access protected route logged out	Request order history with no session	401, no data returned	Critical
TC-044	Token after expiry	Use a token after its 24-hour window	401, forces re-login	High
TC-045	SQL-ish input in login	Enter a quote or SQL fragment in the email field	Treated as invalid input, no injection	High

Orders (TC-051 to TC-065)

TC-ID	Title	Steps	Expected	Priority
TC-051	View own orders	Logged-in user views their order history	200, only that user's orders	Critical
TC-052	Cannot view another user's order	User A requests user B's order by id	403 or 404, never user B's data	Critical
TC-053	Order appears after checkout	Complete a checkout, then view orders	The new order is listed	High
TC-054	Empty order history	New user with no orders views history	200, empty list, not an error	Low
TC-055	Order shows correct total	Open an order placed with three items	Order total matches what was charged	High

Checkout (TC-101 to TC-115)

TC-ID	Title	Steps	Expected	Priority	Notes
TC-101	Checkout with valid discount	Add item, apply SAVE10, submit	Total reduced by 10 percent	Critical	Designed to fail: SAVE10 currently applies 100 percent off (BUG-009)
TC-102	Checkout with empty cart	Submit checkout with no items	400, no order created	High	
TC-103	Checkout with invalid code	Apply code FAKE99, submit	Code ignored, full price charged	Medium	
TC-104	Checkout with out-of-stock item	Add item with stock 0, submit	Order rejected, stock checked first	High	Designed to fail: there is no stock check before checkout (BUG-008)
TC-105	Checkout happy path	Add in-stock item, no code, submit	200, order created, correct total charged	Critical	
TC-106	Discount on already-zero total	Apply SAVE10 to an empty-priced cart	No negative total, handled cleanly	Medium	
TC-107	Double-apply discount	Apply SAVE10 twice in one checkout	Applied once only, not stacked	High	
TC-108	Checkout while logged out	Submit checkout with no session	401, no order created	High	
TC-109	Price tampering	Submit checkout with a modified price in the payload	Server uses its own price, ignores the client value	Critical	
TC-110	Concurrent checkout	Submit the same cart twice in quick succession	One order created, not two	High	

Coverage summary

Area	Cases	Risk priority (Phase 2)
Checkout	10	1 (Critical)
Auth	10	2 (Critical)
Cart	9	3 (High)
Orders	5	2 (Critical)
Products	6	4 (Medium)
Search / sort	6	4 to 5

Total: 46 test cases. The two designed-to-fail cases (TC-101, TC-104) exist to catch BUG-009 and BUG-008 with a written, repeatable check, the same bugs the automated suite encodes in `checkout.spec.ts` .